

RS004 Week 4 — English Worksheet

Topic: Player Faces a Direction (Apply) · **Course:** Platformer Game · **Time:** about 45 minutes

This week your player gets a **face**. A `facing` variable remembers which way it last moved, and a `draw_player()` function uses it to put the eyes on the correct side. One function takes charge of *all* the drawing — that is **separation of concerns**. This worksheet is for thinking about and explaining that code, not for building it here.

Keep these words handy: **facing**, **state variable**, `draw_player()`, **offset**, **separation of concerns**.

1 · Predict 🧠

Read each snippet. Write what you think will happen.

```
facing = "right"
if keys[pygame.K_LEFT]:
    player.x -= PLAYER_SPEED
    facing = "left"
```

The player moves left. What does `facing` become? What if it moves right next?


```
body_color = (220, 50, 50) if facing == "right" else (180, 30, 30)
```

This is a one-line `if` / `else`. What colour is the body when facing left?


```
if facing == "right":  
    eye_x = rect.x + 35  
else:  
    eye_x = rect.x + 15
```

Why is the eye further right (**+35**) when facing right?

2 · Spot the Bug 🐛

Each snippet is broken. Write the fixed code on paper, then explain why the original was wrong.

Bug A — The eyes should follow the player as it moves. Right now the eyes stay frozen in one spot while the body slides away.

```
def draw_player(rect, facing):  
    pygame.draw.rect(screen, (220, 50, 50), rect)  
    pygame.draw.circle(screen, (255, 255, 255), (135, 215), 6)
```

Hint: the eye position is hard-coded as `(135, 215)`. It should be built from `rect.x`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The body colour should change with direction, but it is always the same red.

```
def draw_player(rect, facing):  
    body_color = (220, 50, 50)  
    pygame.draw.rect(screen, body_color, rect)
```

Hint: `facing` is passed in but never used. Make the colour depend on it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — `facing` should update when the player moves, but it never changes from its starting value.

```
keys = pygame.key.get_pressed()
if keys[pygame.K_LEFT]:
    player.x -= PLAYER_SPEED
if keys[pygame.K_RIGHT]:
    player.x += PLAYER_SPEED
```

Hint: each movement should also set `facing`.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Here is a working `draw_player()` that puts the eye on the correct side and changes colour with direction.

```
def draw_player(rect, facing):
    body_color = (220, 50, 50) if facing == "right" else (180, 30, 30)
    pygame.draw.rect(screen, body_color, rect)

    if facing == "right":
        eye_x = rect.x + 35
    else:
        eye_x = rect.x + 15
    eye_y = rect.y + 15
    pygame.draw.circle(screen, (255, 255, 255), (eye_x, eye_y), 6)
```

Read it carefully and answer:

Why does `draw_player()` take `facing` as one of its inputs?

The first line uses a one-line `if / else`. What two colours can `body_color` become, and what decides which one?

The eye uses `rect.x + 35` or `rect.x + 15`. Why is the eye built from `rect.x` instead of a fixed number like `135`?

`eye_y` is always `rect.y + 15`. Why does the up/down position stay the same when the left/right position changes?

This one function draws the body AND the eye. Why is keeping all the drawing in one place (separation of concerns) helpful?

4 · Explain Your Lesson Code 🎥

Explain the code **you wrote in today's lesson**. Record a short video on your phone — you may show your player turning left and right while it runs. Try to use these words: **facing**, **draw_player**, **offset**, **direction**, **function**.

1. Show your player facing right, then left.
2. Point at your `facing` variable and say when it changes.
3. Read your eye-position line out loud and explain the offset.
4. Show one choice you made in your code and why.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.